KAPLAN

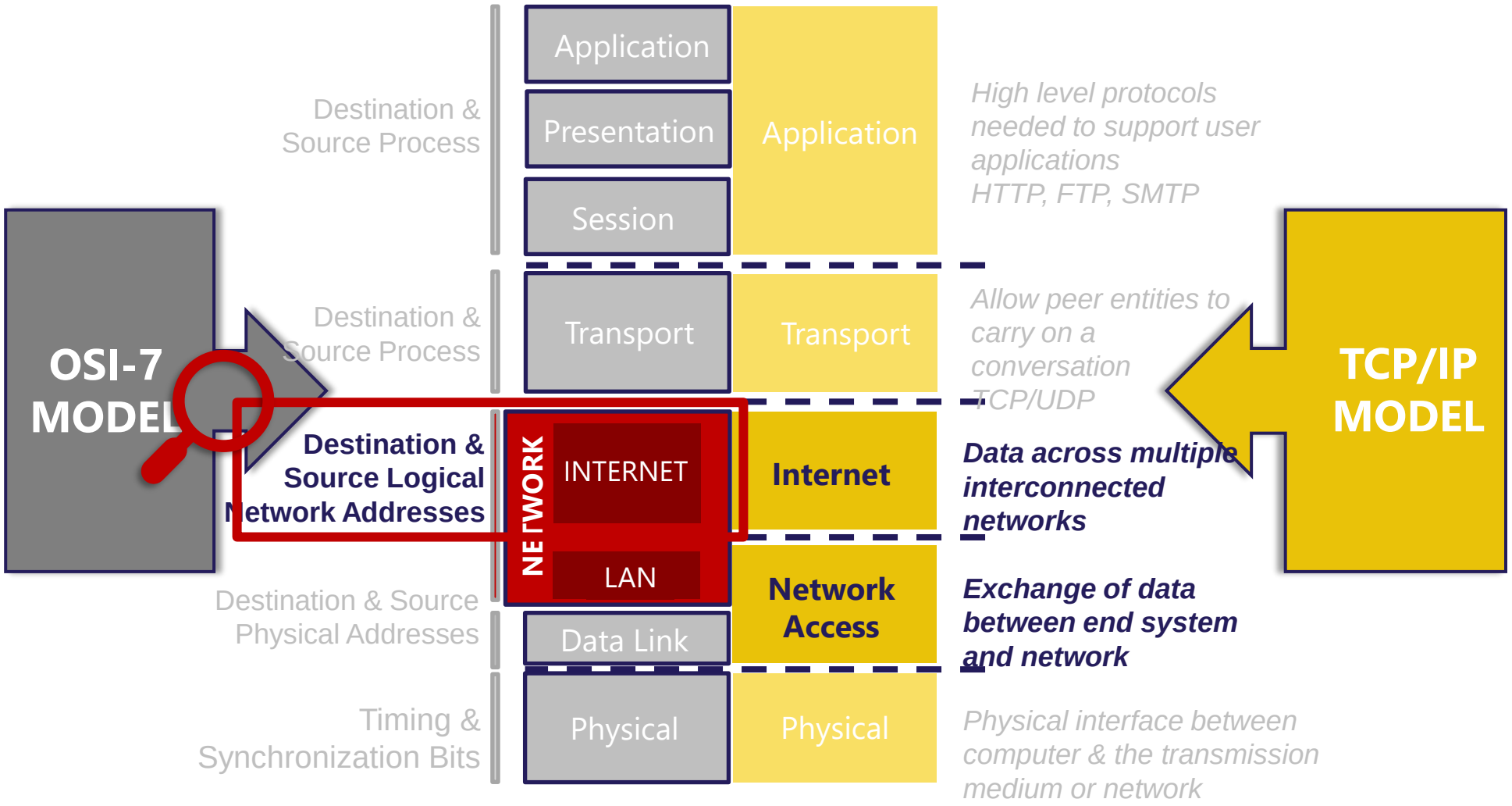
Introduction to Data Communications

Topic 4: Examining the IP Network Layer Protocol, IP Addressing, and Subnetting

Our vision is to be Singapore's private education institution of choice, and we are committed to helping our students achieve their career and educational goals through lifelong learning opportunities.

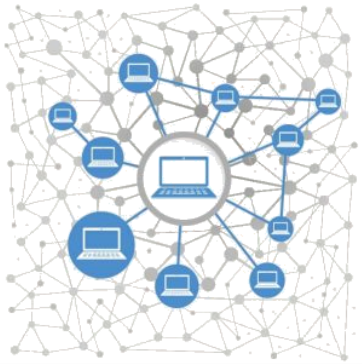
We build futures, one success story at a time.

Integrity . Knowledge . Support . Opportunity . Results


OSI-7 MODEL
Application
Presentation
Session
Application
Destination & Source Process
High level protocols needed to support user applications
HTTP, FTP, SMTP
Transport
Transport
Destination & Source Process
Allow peer entities to carry on a conversation
TCP/UDP
Destination & Source Logical Network Addresses
NETWORK
INTERNET
LAN
Internet
Data across multiple interconnected networks
Network Access
Exchange of data between end system and network
Destination & Source Physical Addresses
Data Link
Timing & Synchronization Bits
Physical
Physical
Physical interface between computer & the transmission medium or network
TCP/IP MODEL

Network Layer



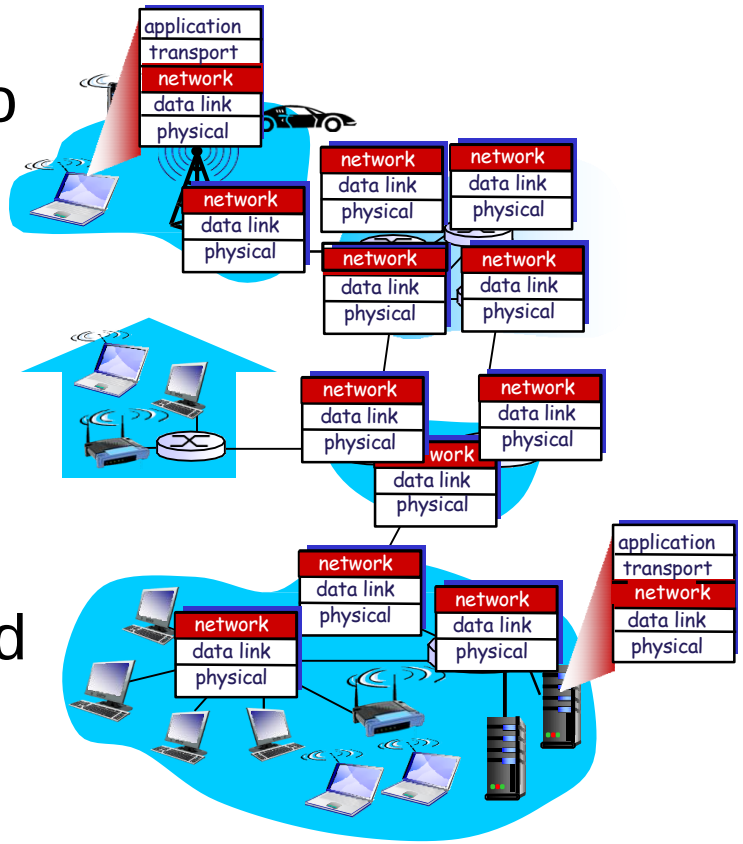


Provides the means of transferring packets from a **source to** a **destination** host via one or more networks

Network Layer

-Transport segment from sending to receiving host

- Sending Side:
  -Encapsulates segments into datagrams
- Receiving Side:
  -Delivers segments to transport layer
- Network layer protocols in every host and router *(layer-3 devices)*



Key Functions

1.Forwarding

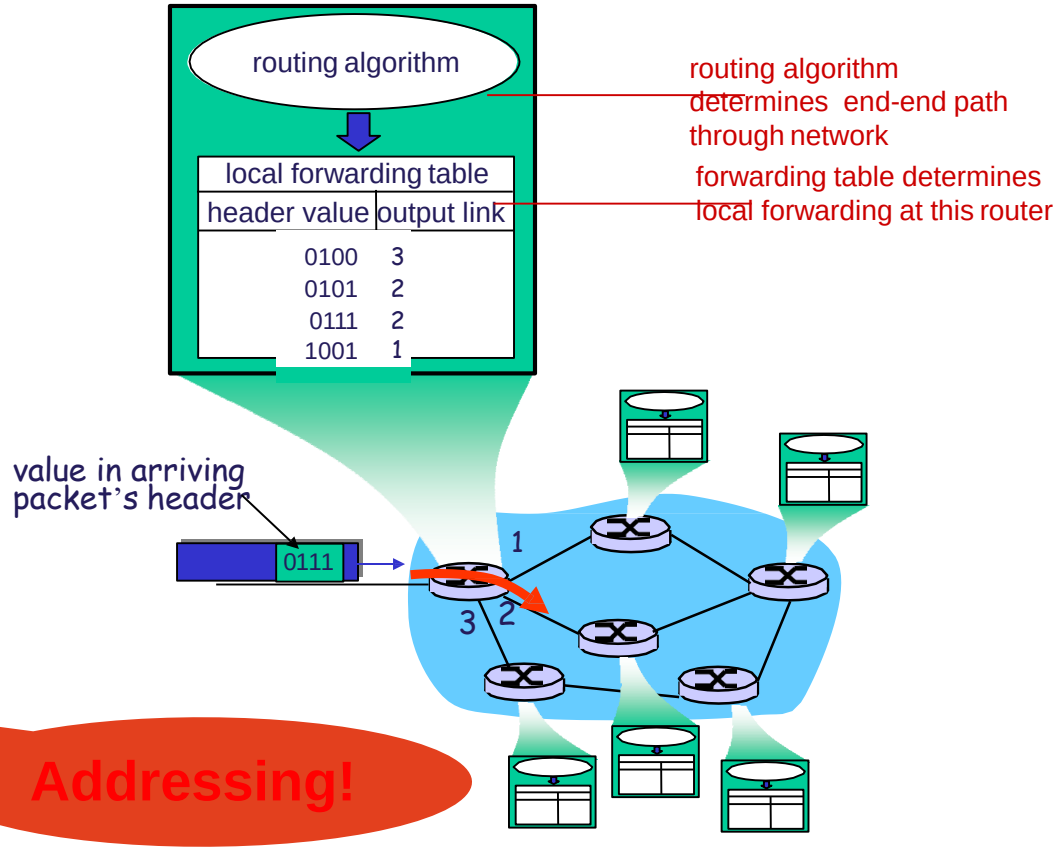
-Move packets from router's input -> appropriate router output

-**Analogy**: the process of getting through the single interchange

2.Routing

-Determine the route taken by packets from source to destination

-**Analogy: the** process of planning trip from source to destination (via *routing algorithms*)

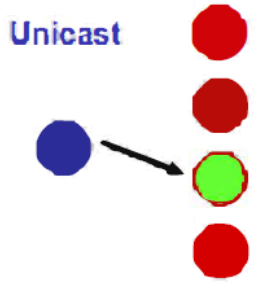


Address Types

- Unicast
- Multicast
- Broadcast
- Anycast
- Geocast

Unicast

- Each unicast client that links to the server takes up extra bandwidth
- **Client** has a **direct relationship** to the **server**

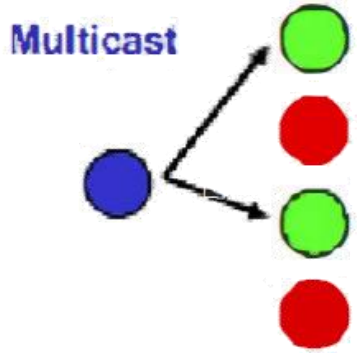


Unicast Messages- The same number of messages have to be sent as many as receivers. If there are 3 recipients, then 3 messages have to be sent. Think about it. What if you have 100 recipients? This will be **bandwidth intensive**.

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Multicast

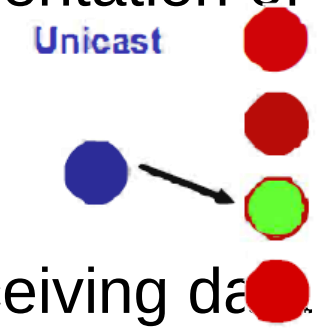
- A multicasting is associated with a group of **interested receivers**
- **No direct relationship** between the clients and server
- Clients are connected to the multicast address, **no additional overhead** on the server.
  -i.e. server sends out only one stream per multicast station. same load is experienced on the server whether only one client or 1,000 clients are listening
- Can be used across a **WAN**

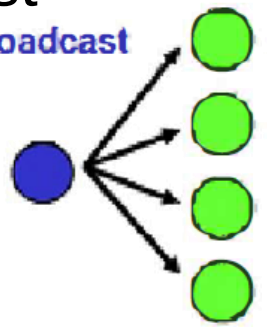


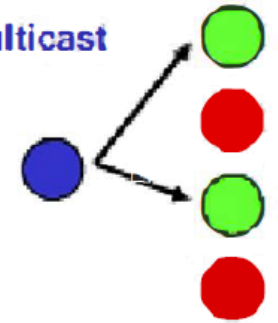
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Multicast / Broadcast

- **One-to-many** communications
- IP Multicasting refers to the implementation of multicast communication on the internet
- **Multicast is driven by receivers:**
  -Receivers indicate interest in receiving da







Broadcast Message, this time all the network nodes even if they do not need that message will receive the message. This is an **unnecessary traffic**.

Anycast [RFC: 1546]

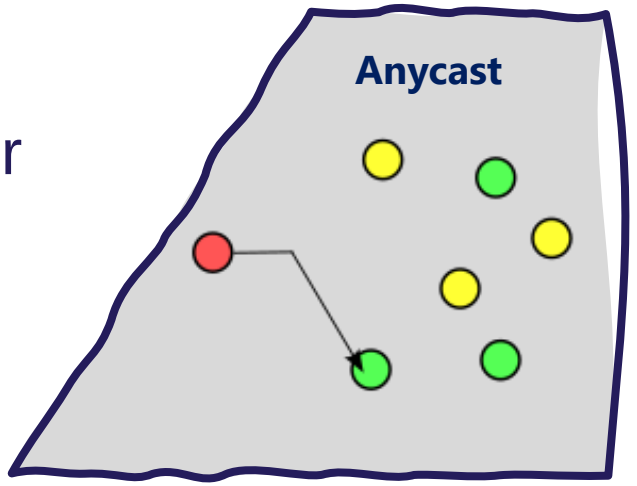
- Anycast addressing is a **one-to-one-of-many** association where datagrams are routed to any single member of a group of potential receivers **that are all identified by the same destination address**

- The routing algorithm selects the single receiver from the group **based on** the **least-expensive routing metric**

  -i.e. the packets are routed to the topologically nearest member of an anycast group

  - **Widely used** for Content Delivery Network **(CDN)** products to bring their content closer to the end user

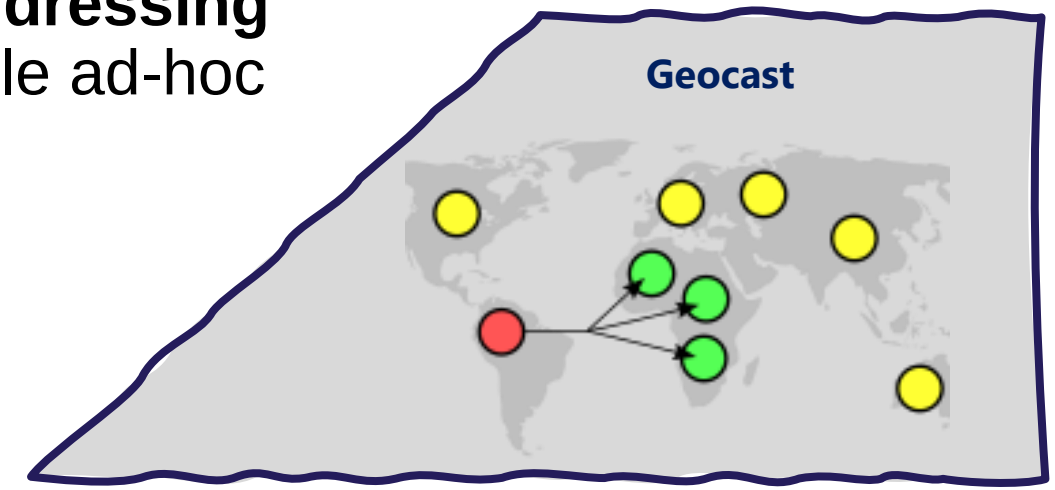
Anycast Message that the packet is sent to the closest host between the destinations that has the same IP address.



Geocast

- Geocast refers to the delivery of information to a group of **destinations in a network identified by their geographical locations**

- It is a **specialized form of multicast addressing** used by some routing protocols for mobile ad-hoc networks.



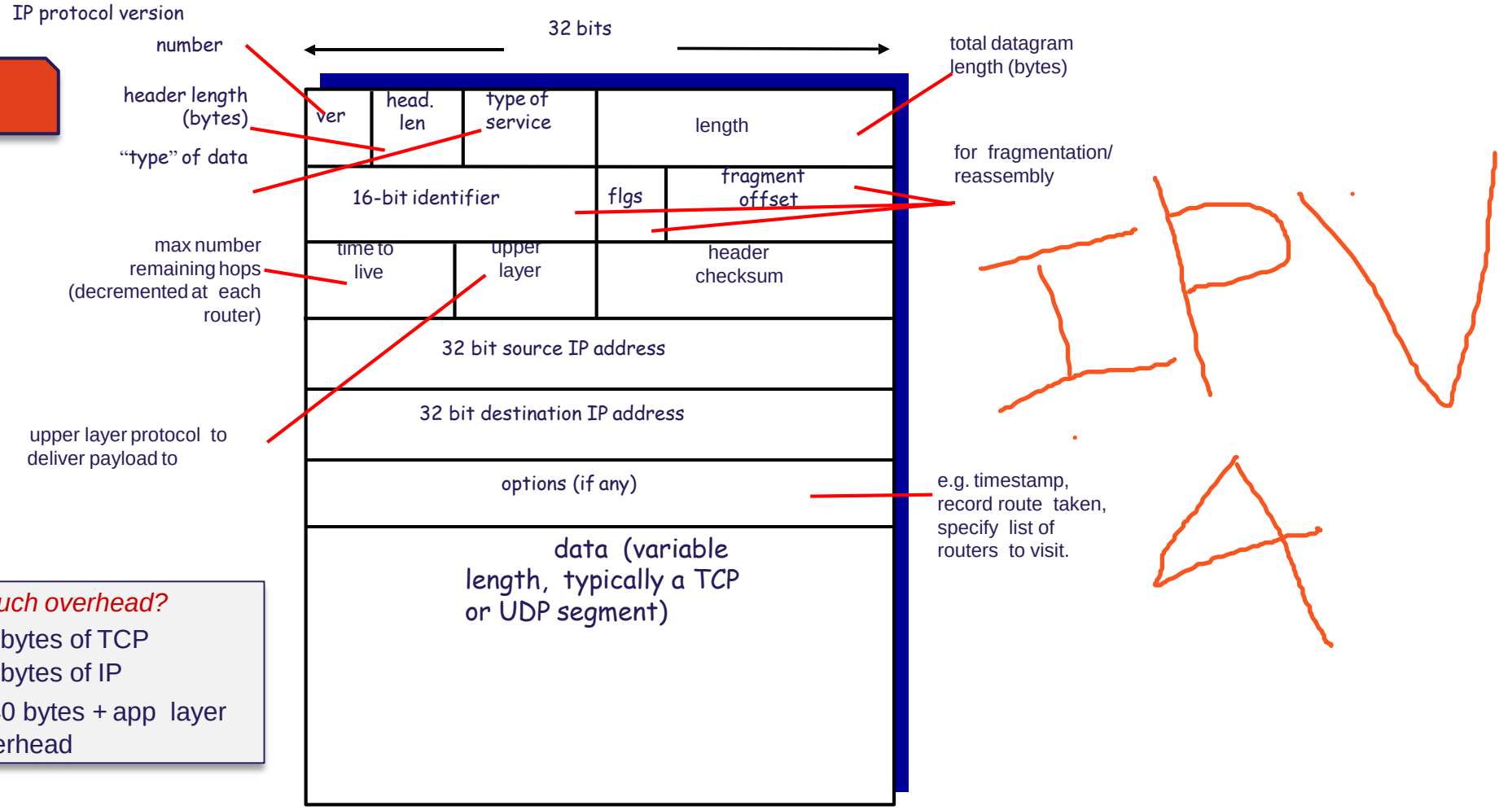
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IPv4 Addressing

- Fundamentals
- Subnets
- Classful, Classless (CIDR) IP
- Static vs Dynamic IP
- Obtaining a Global IP Address
  - -ISP Address Allocation
  - -Hierarchical Addressing
  - -ICANN
- Special IPv4 Addresses

IP Datagram

Header

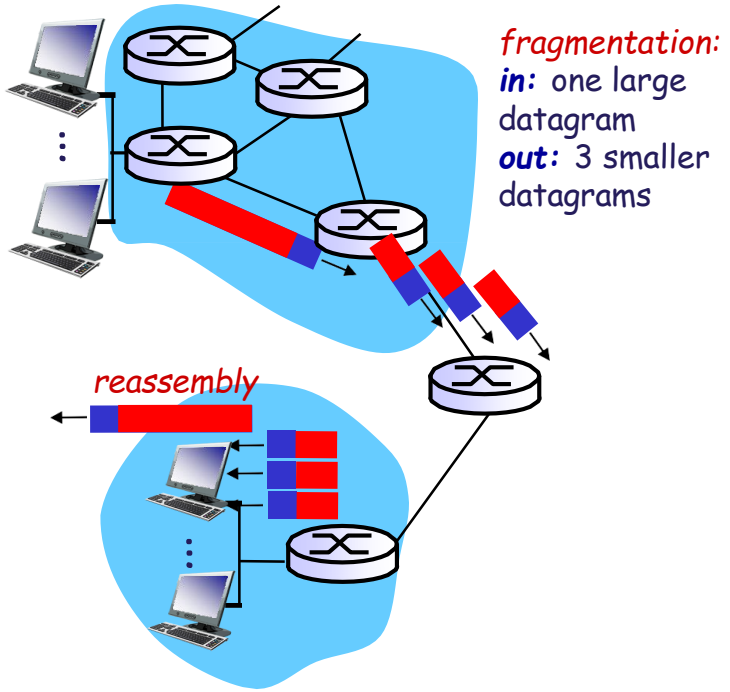


how much overhead?

- 20 bytes of TCP
- 20 bytes of IP
- = 40 bytes + app layer overhead

IP Fragmentation, Reassembly

- Network links have **MTU** (Max Transfer Size)
  - -Largest possible link-level frame
  - -Different link types, different MTUs
- **Fragmentation:** Large IP datagram divided within the network
  - -One datagram becomes several datagrams
- **Reassembly:** Only at final destination
  - -IP header bits used to identify & order related fragments

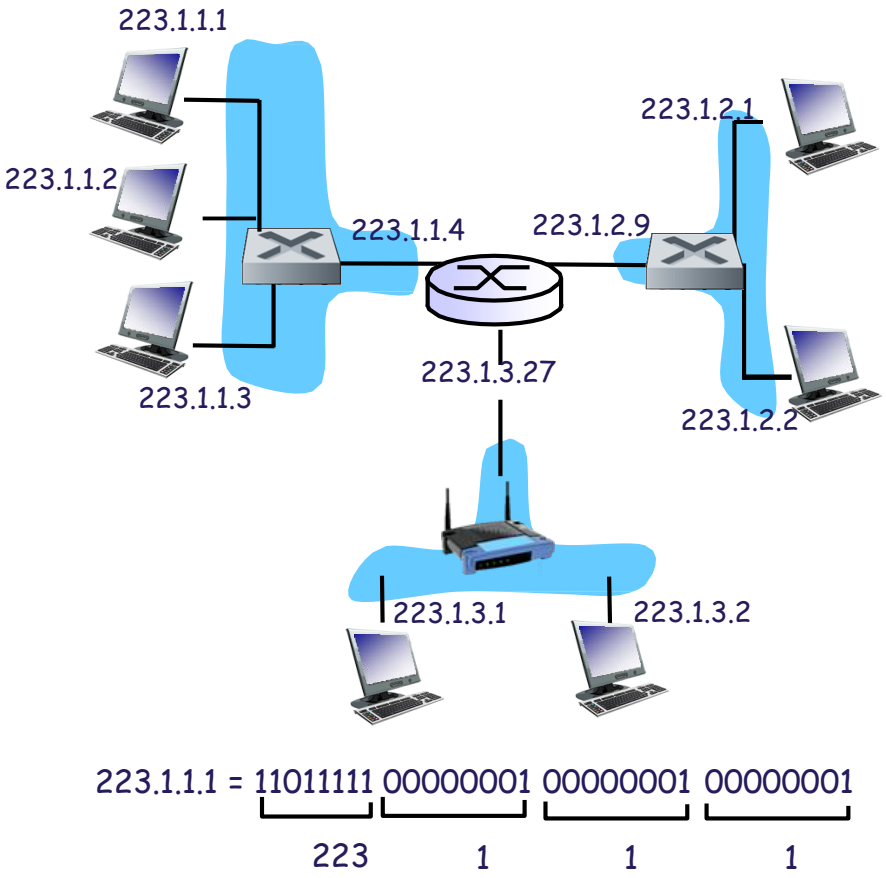


IP Addressing

- **IP address:** 32-bit identifier for host and router interface
- **Interface:** connection between host/router and physical link
  - -**Routers:** multiple interfaces
  - -**Host:** one or two interfaces

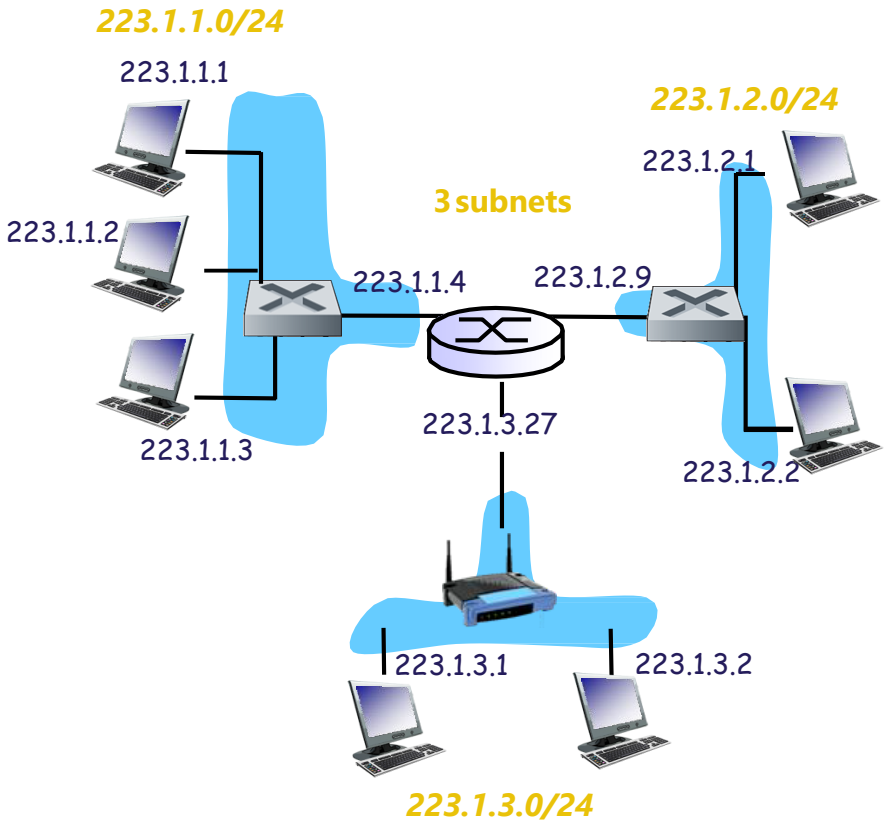
(e.g., wired Ethernet, wireless 802.11)

“IP addresses for each interface”



Subnets

- A **subnetwork** or **subnet** is a logical subdivision of an IP network
- Isolated networks with **Subnet ID**
  - Can physically reach each other in same sub-network without intervening router
- **IP address:**
  - **Subnet ID** - high order bits
  - Host ID - low order bits



223.1.1.1/24 = 11011111 00000001 00000001 00000001

223 1 1 1

Subnet Mask: /24

IP Address Types

- **Classful**

2*N-2(N=number of host bit)

Class A=1-126_A
Class B=128-191-B
Class C=192-223-C

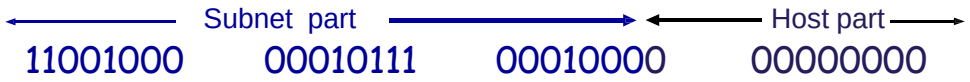
Class	Start bits	No. of network bits	No. of host bits	Range	
A	**0**	**8**	**24**	**1.0.0.1** to **126.255.255.254**	Supports 16 million hosts on each of 127 networks.
B	**10**	**16**	**16**	**128.1.0.1** to **191.255.255.254**	Supports 65,000 hosts on each of 16,000 networks.
C	**110**	**24**	**8**	**192.0.1.1** to **223.255.254.254**	Supports 254 hosts on each of 2 million networks.
D	**1110**	**undef**	**undef**	**224.0.0.0** to **239.255.255.255**	Reserved for multicast groups.
E	**1111**	**undef**	**undef**	**240.0.0.0** to **254.255.255.254**	Reserved for future use, or research and development purposes.

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IP Address Types

- **CIDR**: **C**lassless **I**nter**D**omain **R**outing
  - Subnet portion of address of arbitrary length
  - Address format: a.b.c.d/x, where x is no. of bits in subnet portion of address
  - Possible to split host part further to create additional subnetting

23 belong to N
9 belong to H



200.23.16.0/23
Subnet mask is /23 or
255.255.254.0

Subnet Mask example

- **IP address 192.168.40.28/21 - what's the network ID?**
  - Subnet Mask /21: 255.255.248.0
  - 40 = 00101000 with mask 11111000
  - Network ID is 192.168.40.0 /21
- **192.168.45.52 /21 - what's the network ID?**
  - Subnet Mask /21: 255.255.248.0
  - 45 = 00101101 with mask 11111000
  - Network ID is 192.168.40.0 /21

Type text here

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Static vs Dynamic IP Addresses

- **Statically Assigned**
  - Hard-coded by system admin in a file
    - **Windows:** control-panel->network->configuration->tcp/ip->properties
    - **UNIX:** /etc/rc.config

- **Dynamically Assigned**
  - **DHCP:** Dynamic Host Configuration Protocol
  - Dynamically get address from a server (or router)
    - Temporarily assigned, or "leased"
    - After a period of time, this lease "expires,"
      - Renews your old address or assigns new IP address

ICANN

- How does an ISP get block of addresses?

- **ICANN**: **I**nternet **C**orporation for **A**ssigned **N**ames and **N**umbers
  http://www.icann.org/

  -Allocates addresses

  -Manages DNS

  -Assigns domain names, resolves disputes

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Why IP and MAC both?

- **MAC Address Only**
  - -MAC/Physical address is a **globally unique ID** for your device
  - -Useful and efficient for **local communications (LAN)**
  - -Broadcast domain – the **internet is not possible**
  - -MAC address is burnt with hardware NIC card – loss of hardware, loss of all existing connections
- **IP Address Only**
  - -They can **group and organize** different networks (hierarchical)
  - -They are much like your mailing address
  - -They are **flexible and changeable** – making a device mobile

*tells **who** you are*

00:1C:A2:01:A3:45
00:A0:C9:14:C8:29
00:1B:44:11:3A:B7

*tells **where** you are*



Private IP Addressing

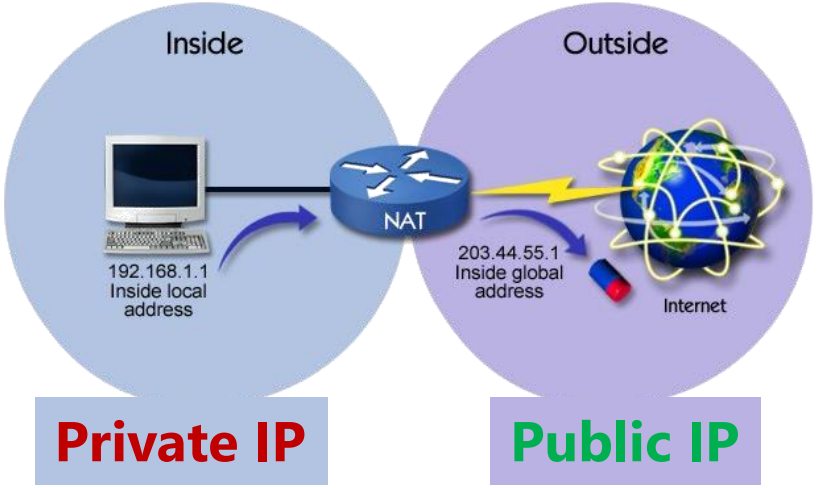
- Fundamentals
- Network Address Translation
  - -SNAT
  - -DNAT
  - -PAT
  - -Port Forwarding

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Private IP Addressing

- **What are private IPs?**
  - **Non-routable** addresses **in internet**
    - -Routers would not deliver packets with private IP addresses
  - **Free to use** without anyone's permission

- **Why do we need them?**
  - There are only 4.3 billion IPv4 addresses
  - They have extended IPv4's life



Private IP Addressing – IPv4

Class A	10.0.0.0 to 10.255.255.255	10.0.0.0/8
Class B	172.16.0.0 to 172.31.255.255	172.16.0.0/12
Class C	192.168.0.0 to 192.168.255.255	192.168.0.0/16

Only a portion

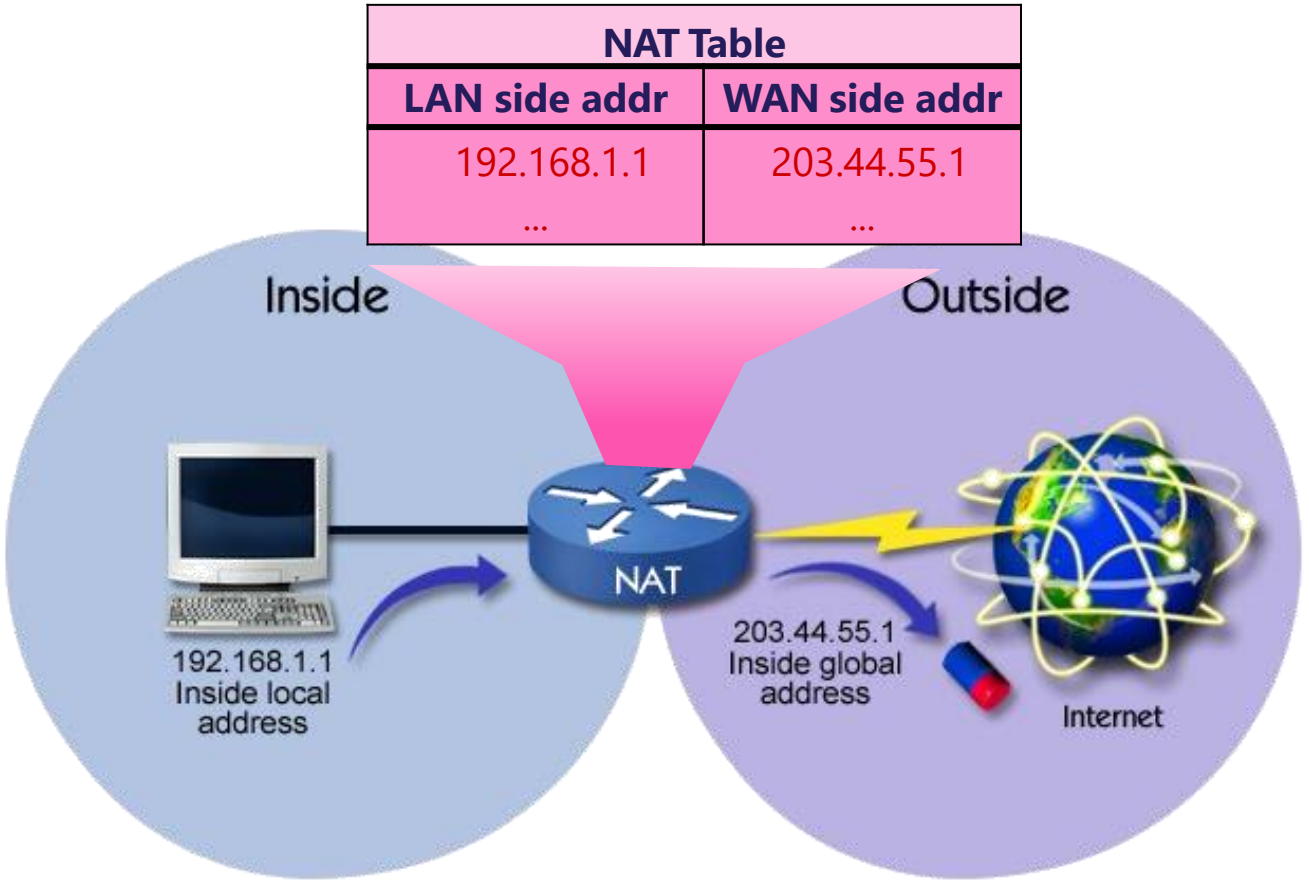
Devices with **private IP** address **cannot connect directly to the Internet**

Instead, **access** to the **Internet** must be brokered by a **router** that **supports Network Address Translation (NAT)**

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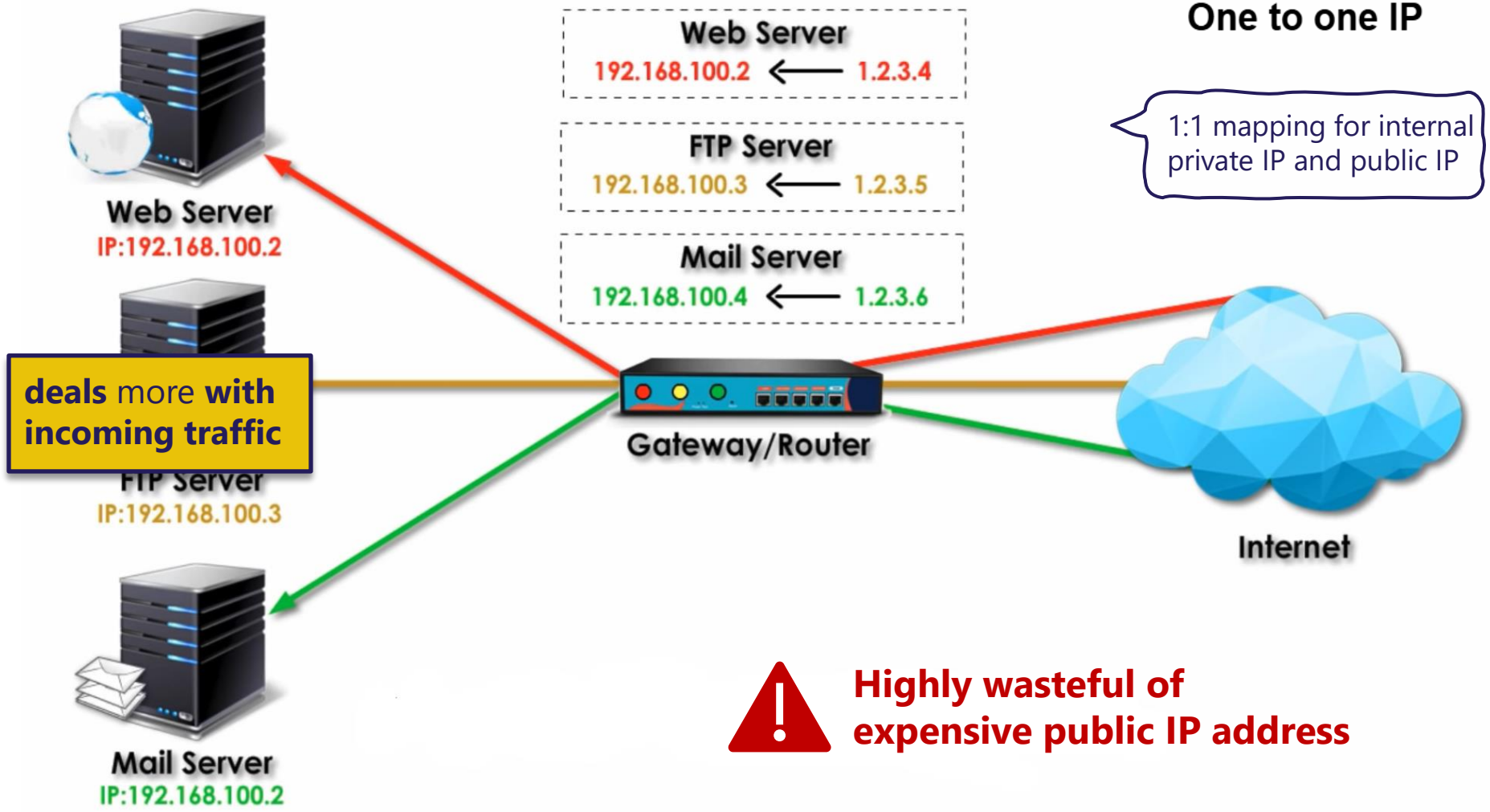
What is NAT?

Motivation: local network uses just one IP address as far as outside world is concerned



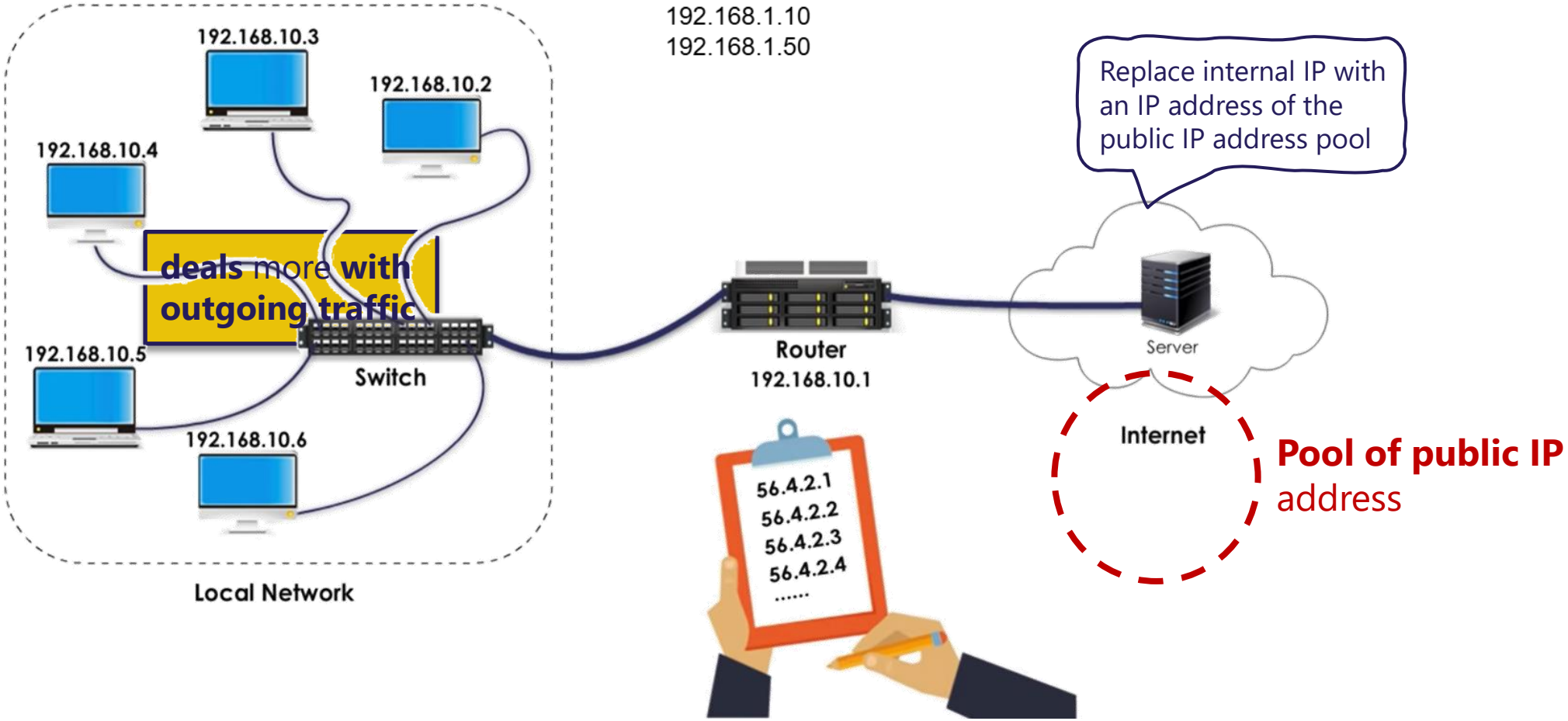
A router **translates an internal host's private IP** address **into** its **public IP address** for outgoing/incoming traffic and vice-versa

SNAT: Static NAT



Highly wasteful of expensive public IP address

DNAT: Dynamic NAT



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IPv6

- Fundamentals
- Address Space
- IPv6 Datagram
- IPv6 Address Simplification
- Multicast addressing
  - -Solicited node address
- Unicast addressing
  - -Global Unicast Address

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Introduction to IPv6

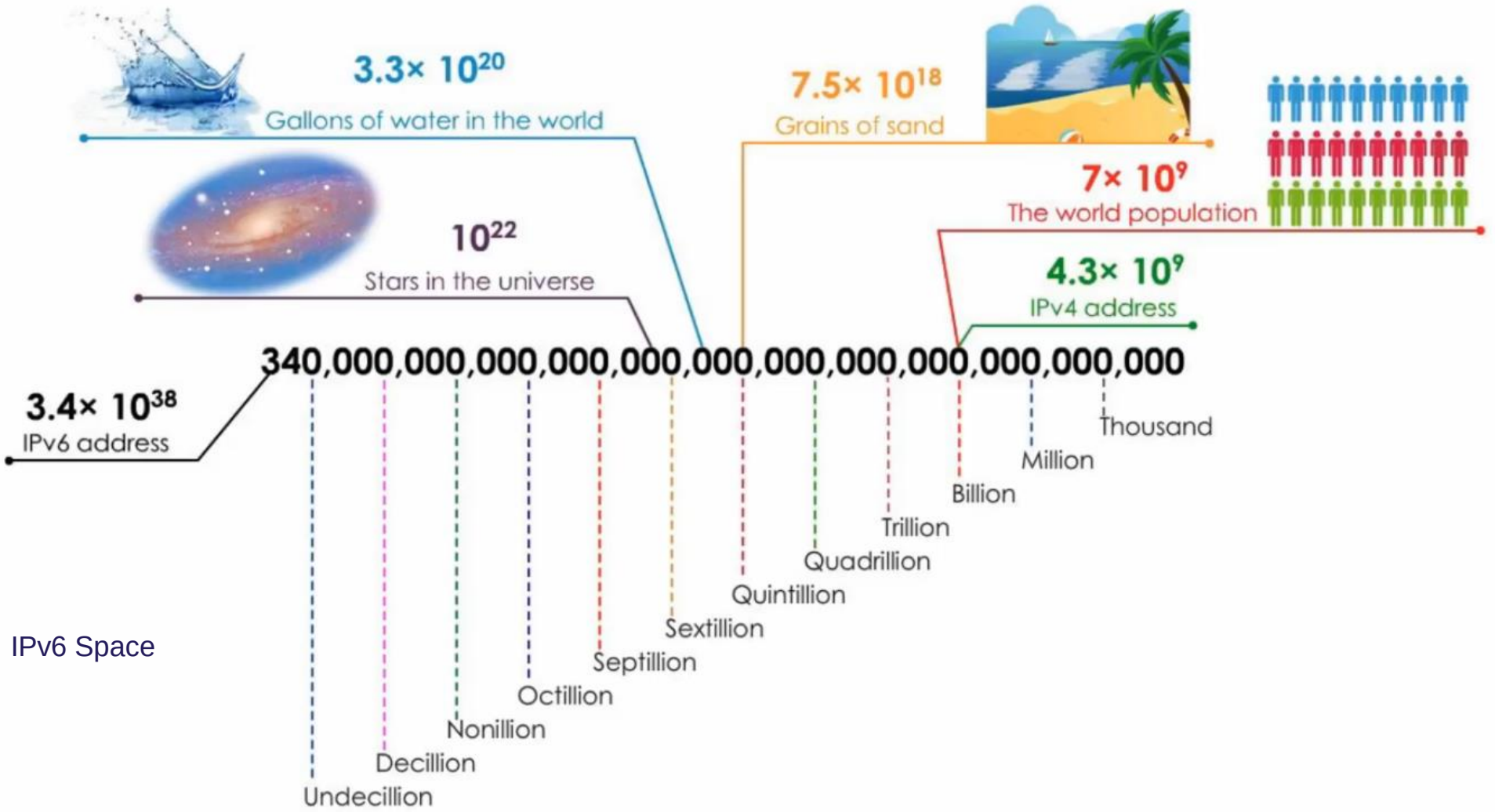
Initial Motivation:

32-bit address space soon to be completely allocated

- Why IPv6?
  - Supports 2^{128} ~= 3.4 x 10^{38} addresses

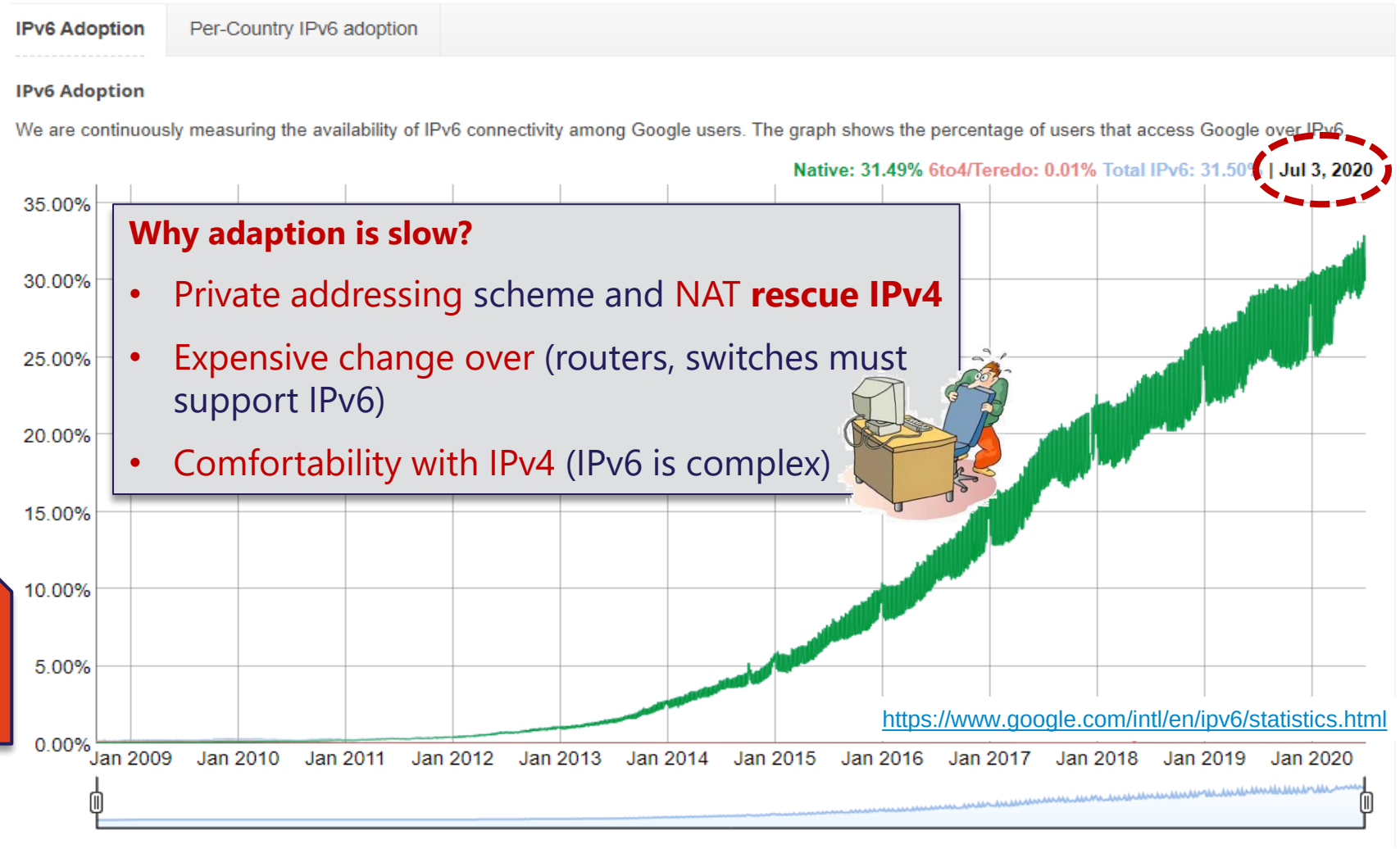
`i.e. 2001:0000:0000:0000:0080:ACDE:02CE:1234`

  - header format helps speed processing/forwarding
  - header changes to facilitate **QoS**


3.3× 10^20
Gallons of water in the world
7.5× 10^18
Grains of sand
7× 10^9
The world population
10^22
Stars in the universe
4.3× 10^9
IPv4 address
340,000,000,000,000,000,000,000,000,000,000,000
3.4× 10^38
IPv6 address
Thousand
Million
Billion
Trillion
Quadrillion
Quintillion
Sextillion
Septillion
Octillion
Nonillion
Decillion
Undecillion

IPv6 Space

IPv6 Last Words



IPv4 to IPv6 Address Transition

- Dual Stack
- Tunnelling
- NAT64 Translation

IPv4 to IPv6 Transition

- IPv4 and IPv6 will coexist for a long time
- Techniques

1. Dual Stack

A dual-stacked device (PC, Server, Router) supports both IPv4 and IPv6

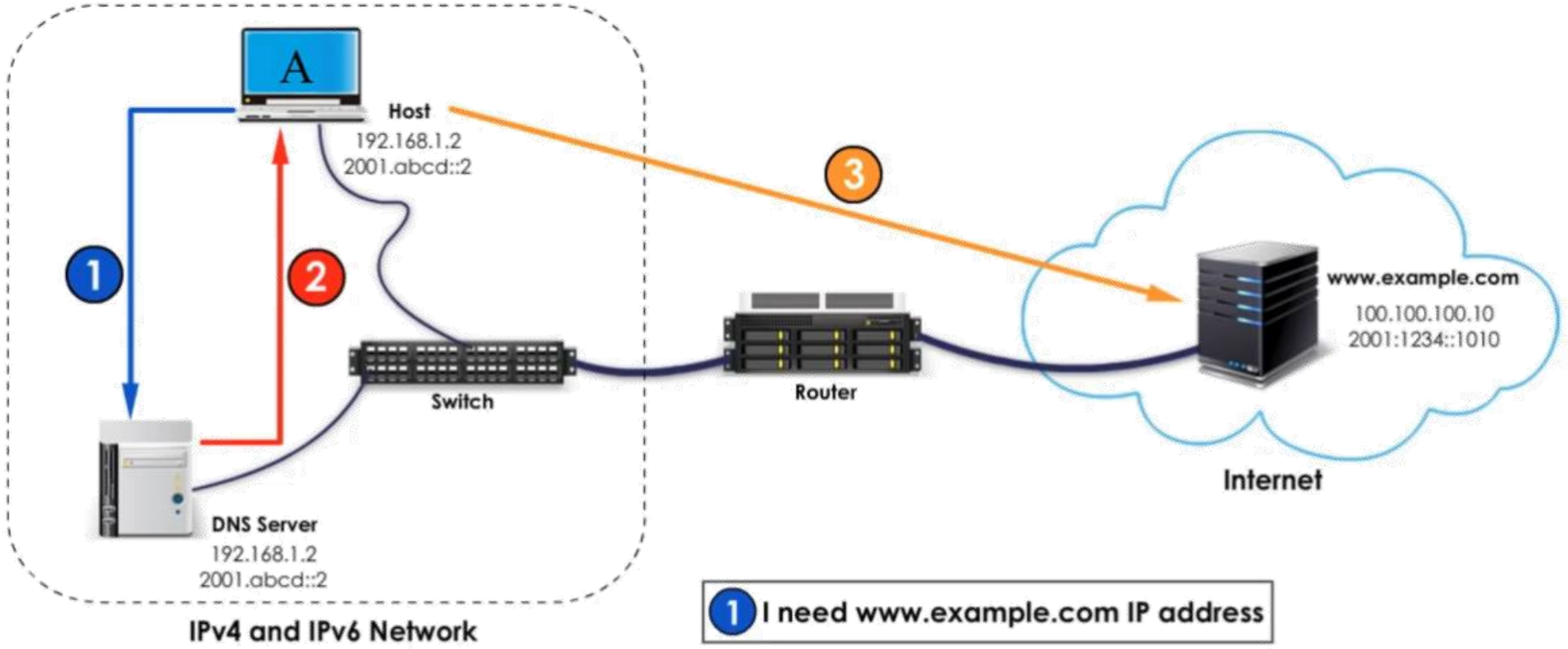
2. Tunneling

1. **Manual Tunneling**
2. **6to4**
3. **ISATAP** (Intra Site Automatic Tunnel Addressing Protocol)

3. Translation (NAT64)

Similar to address translation

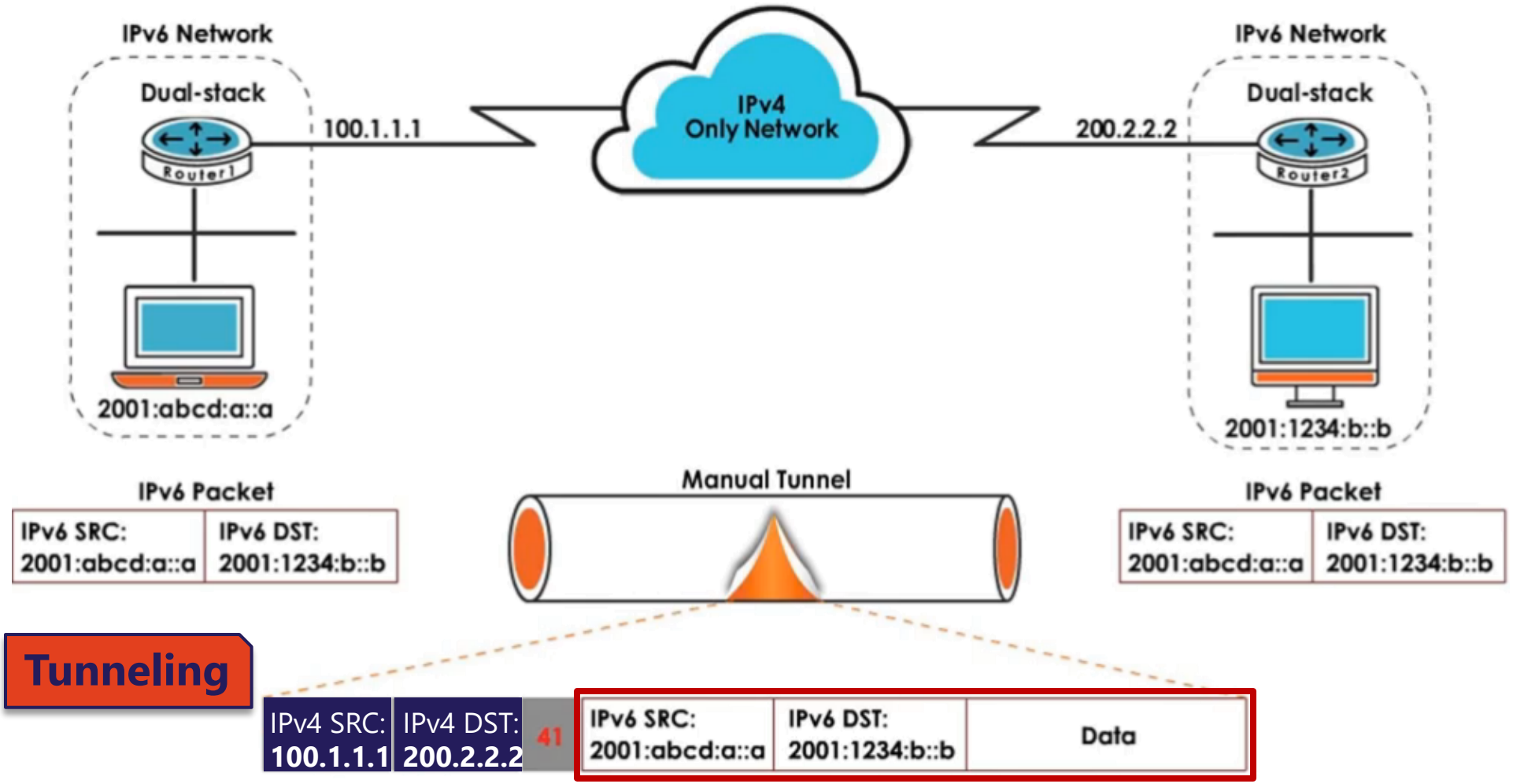
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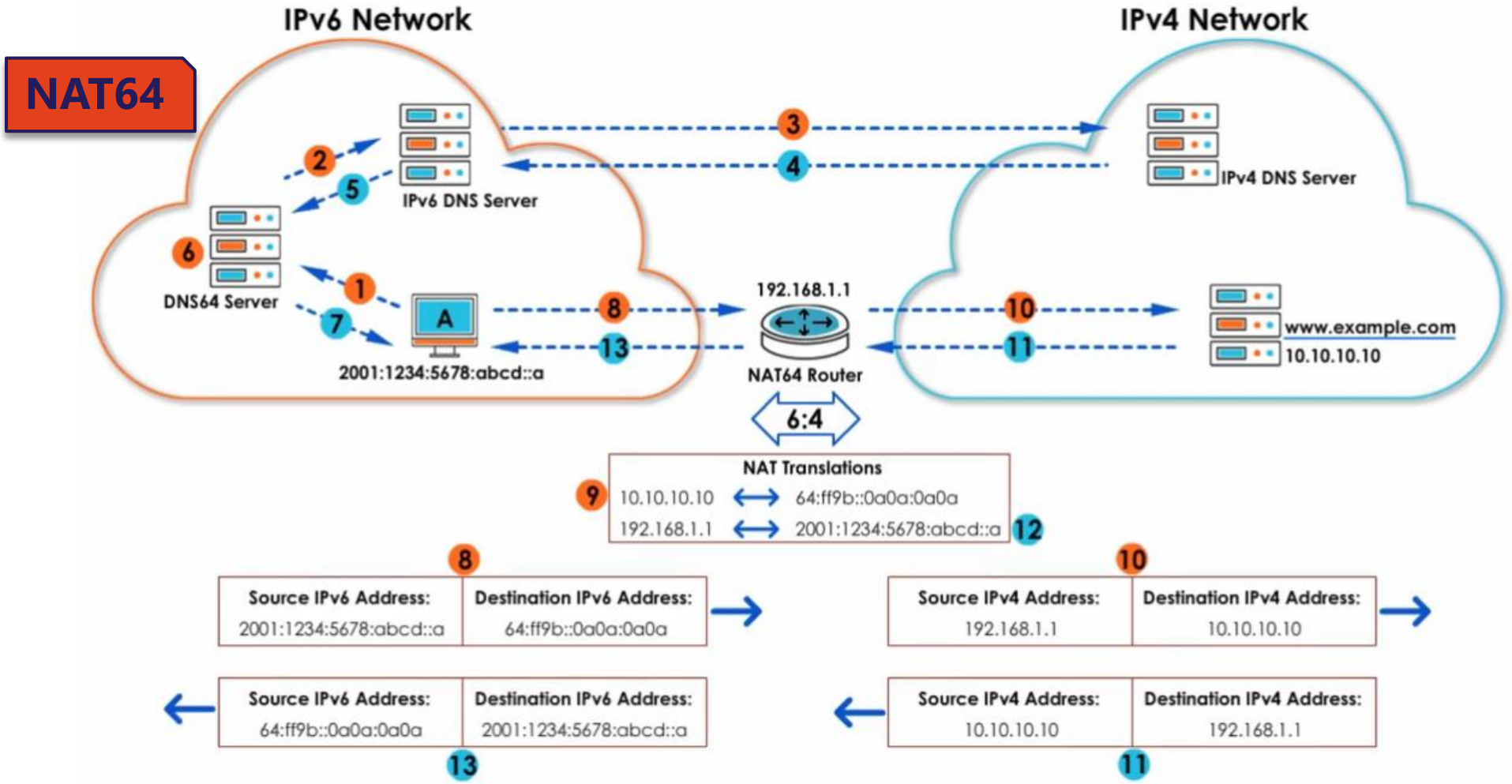
Dual Stack

1. I need www.example.com IP address
2. Type AAAA record: 2001:1234::1010
   Type A record: 100.100.100.10
3. IPv6 Session with 2001:1234::1010

If failed, IPv4 Session will be established


IPv6 Network
Dual-stack
Router1
100.1.1.1
IPv4
Only Network
200.2.2.2
IPv6 Network
Dual-stack
Router2
2001:abcd:a::a
2001:1234:b::b
IPv6 Packet
IPv6 SRC:
2001:abcd:a::a
IPv6 DST:
2001:1234:b::b
Manual Tunnel
IPv6 Packet
IPv6 SRC:
2001:abcd:a::a
IPv6 DST:
2001:1234:b::b
Tunneling
IPv4 SRC:
100.1.1.1
IPv4 DST:
200.2.2.2
41
IPv6 SRC:
2001:abcd:a::a
IPv6 DST:
2001:1234:b::b
Data
Payload – IPv6 Packet

NAT64


IPv6 Network
IPv4 Network
IPv6 DNS Server
IPv4 DNS Server
DNS64 Server
2001:1234:5678:abcd::a
192.168.1.1
NAT64 Router
www.example.com
10.10.10.10
6:4
NAT Translations
10.10.10.10 ⟷ 64:ff9b::0a0a:0a0a
192.168.1.1 ⟷ 2001:1234:5678:abcd::a
Source IPv6 Address: 2001:1234:5678:abcd::a
Destination IPv6 Address: 64:ff9b::0a0a:0a0a
Source IPv4 Address: 192.168.1.1
Destination IPv4 Address: 10.10.10.10
Source IPv6 Address: 64:ff9b::0a0a:0a0a
Destination IPv6 Address: 2001:1234:5678:abcd::a
Source IPv4 Address: 10.10.10.10
Destination IPv4 Address: 192.168.1.1

ICMP: Internet Control Message Protocol

- Used by hosts & routers to communicate network-level information
  - **error reporting:** unreachable host, network, port, protocol
  - echo request/reply (used by **ping**)
- **ICMP message**
  - type, code plus first 8 bytes of IP datagram causing error
- **ICMPv6: (IPv6)**
  - Additional message types, e.g. “Packet Too Big”
  - Multicast group management functions



Type	Code	description
0	0	echo reply (ping)
3	0	dest. network unreachable
3	1	dest host unreachable
3	2	dest protocol unreachable
3	3	dest port unreachable
3	6	dest network unknown
3	7	dest host unknown
4	0	source quench (congestion control - not used)
8	0	echo request (ping)
9	0	route advertisement
10	0	router discovery
11	0	TTL expired
12	0	bad IP header

ARP: Address Resolution Protocol

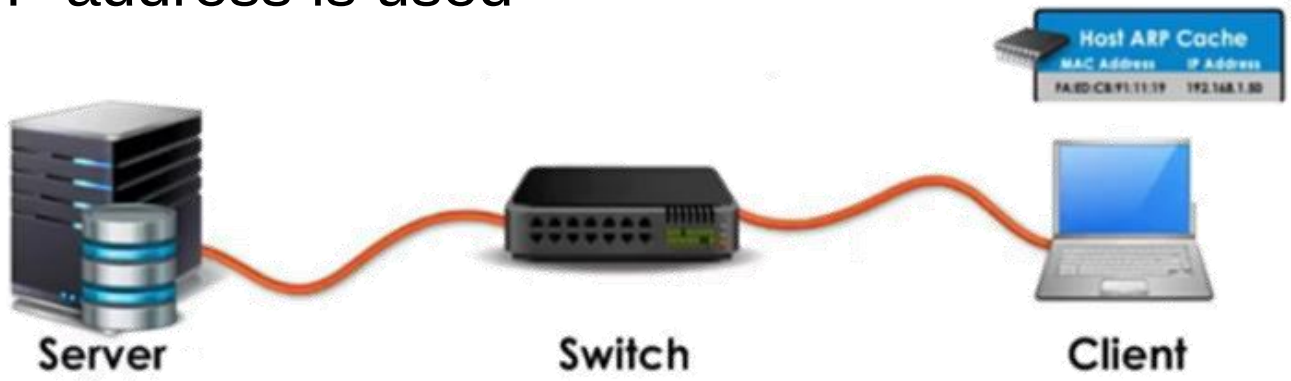
-**Maps an IP Address** to a physical **MAC Address on a LAN**

- A computer uses **ARP program** to find another computer's MAC address based on its IP address

LAYER 2 PROTOCOL

- Why MAC address?
  - **Communication within LAN:** MAC address is used
  - **Communication between LANs:** IP address is used

- **ARP cache:**
  - A table of IP address with their corresponding MAC addresses



Thank you

